

Enhanced Image Inpainting using Transformer–GAN Hybrid Models

A Comprehensive Overview

Introduction and Background

Image inpainting (also known as image completion) is the process of filling in missing or corrupted regions of an image with plausible content that seamlessly blends with the surrounding area. This problem has a rich history in computer vision. Early **traditional inpainting methods** (pre-deep-learning) were often based on diffusion or patch matching. For example, Bertalmio *et al.* (2000) introduced a diffusion-based approach that propagates neighboring pixel intensities into holes, and later works like Telea (2004) improved efficiency of this process. Exemplar-based methods such as Criminisi *et al.* (2004) search for similar patches in the image and copy them into the missing regions, preserving texture details. These classical methods work well for small gaps or simple textures but struggle with larger holes or complex structures, due to the lack of high-level understanding.

The advent of deep learning brought **learning-based inpainting** methods that significantly improved results by understanding semantic context. Early deep approaches like Pathak *et al.* 2016, *Context Encoders* trained an encoder–decoder convolutional network to predict missing center regions of an image, using both reconstruction loss and an adversarial loss to ensure realism. This was one of the first applications of **Generative Adversarial Networks (GANs)** to inpainting, demonstrating that neural networks can learn to hallucinate plausible content. Building on this, Iizuka *et al.* 2017, *Globally and Locally Consistent Image Completion* introduced the idea of using two discriminators: a global discriminator to enforce overall image realism and a local discriminator focusing on the filled region for finer detail consistency. Their generator network employed dilated convolutions to increase the receptive field, an early attempt to capture more global context.

By 2018–2019, a number of influential inpainting models had emerged. Yu *et al.* 2018, *Generative Image Inpainting with Contextual Attention* proposed a two-stage network that first predicts a coarse fill and then refines it with a *contextual attention* module, which explicitly searches for relevant patches in the unmasked regions of the image to borrow details from. This attention mechanism helped ensure that repeated or symmetric patterns (like bricks or tiles) in the known regions continued correctly into the filled area. Liu *et al.* 2018, *Partial Convolutions for Irregular Holes* introduced a masked convolution operation (partial conv) to handle irregular mask shapes, updating only valid pixels at each layer – an approach that improved results when holes are random and not just centered or rectangular. Similarly, Yu *et al.* 2019, *Free-Form Image Inpainting with Gated Convolution* generalized inpainting to arbitrary masks by learning a gating mechanism on convolutional

filters, dynamically selecting features useful for filling holes. Around the same time, others explored structural guidance: e.g. Nazeri *et al.* 2019, *EdgeConnect* performed inpainting in two stages by first predicting edges for the missing region and then using those edges to guide image completion, which is especially effective for preserving object boundaries and layouts. These foundational deep-learning approaches set the stage by demonstrating the importance of both local texture synthesis and global structure coherence.

Despite these advances, convolutional neural network (CNN) based models have intrinsic limitations for large missing areas due to finite receptive fields. Even with dilated or partial convolutions, purely CNN models may struggle to capture very long-range dependencies or interactions between distant parts of an image. This often led to blurry or inconsistent results when the hole is large or when the image has complex, non-repetitive structures that require a holistic understanding of the scene.

In summary, key foundational concepts in image inpainting include:

- **Maintaining Texture and Structure:** Successful inpainting must produce both realistic local textures (e.g. continue the pattern of bricks or fur) and globally consistent structures (e.g. complete a person's arm in a natural position). Early deep models addressed texture with CNNs and adversarial losses, and gradually incorporated explicit handling of structure via attention or separate predictions (edges, segmentation, etc.).
- **Adversarial Training:** The use of GANs introduced a discriminator network to distinguish real vs. inpainted images. This encourages the generator to produce more photorealistic outputs beyond what pixel-wise losses achieve, helping to reduce blur and unrealistic artifacts. Many inpainting models combine a reconstruction loss (like L_1 or L_2) with one or more adversarial losses.
- **Two-Stage or Multi-Stage Models:** A common strategy is to have a coarse network fill in a rough guess of the missing content, followed by a refinement network to add fine details. This divide-and-conquer approach can stabilize training and improve quality. The refinement stage often incorporates attention or uses high-frequency guidance (like predicted edges or gradients).
- **Mask Handling:** The irregular and arbitrary shape of missing regions led to architectural innovations like partial and gated convolutions, as well as the use of mask channels or mask-specific normalization layers to inform the network which pixels are known vs. unknown at each layer. Proper mask handling prevents naive artifacts (such as simply blurring edges of holes) and ensures new content is generated rather than trivially copied.

These foundational ideas serve as a basis for more recent developments that aim to further enhance inpainting quality by employing advanced architectures like Transformers and by combining them with GAN frameworks. In the following sections, we delve into the role of GANs and Transformers in vision, and how *hybrid Transformer-GAN models* push the state-of-the-art in image inpainting.

GANs for Image Inpainting and Vision Generation

Generative Adversarial Networks (GANs) are a class of generative models introduced by Goodfellow *et al.* in 2014. A GAN consists of two networks trained in opposition: a **generator** G that tries to synthesize realistic data (in our case, completed images), and a **discriminator** D that tries to distinguish between real images and the generator's outputs. The adversarial training objective drives G to produce outputs that D cannot reliably tell apart from true images.

In image inpainting, GANs are typically used in a *conditional* setting: the generator is conditioned on the partially observed image (with holes) and outputs a completed image. The discriminator may also be conditioned on the image with holes, or it may see the completed image alone and try to judge if it's a real, fully intact image. GANs are crucial for producing sharper and more realistic textures:

- **Global vs. Local Discriminators:** As introduced by Iizuka *et al.*, using two discriminators at different scales helps enforce both global coherence and local detail. The *global discriminator* looks at the entire image and ensures that the completed image overall looks natural (for example, preventing an inpainted face from having two different eye colors or the sky on one side of an image from being a different time of day than the other). The *local discriminator* focuses on the region of the fill (usually by cropping a region around the hole) to ensure high-frequency details in the filled area match the surrounding context.
- **Patch-Based Discriminators:** Many inpainting models employ a PatchGAN discriminator (which classifies small patches of the image as real/fake instead of the whole image) to encourage the generator to get local texture right. This technique, borrowed from image-to-image translation, effectively models textures and is less heavy than a full-image discriminator.
- **Adversarial Loss Weighting:** Because GAN training can be unstable, inpainting models often balance the adversarial loss with reconstruction losses. Early in training, a higher weight on L_1 or L_2 loss helps the network learn the coarse structure; later, the adversarial loss (and possibly a perceptual loss) help refine realism. This balance is important to prevent common GAN artifacts or mode-collapse where the network might fill unrelated but visually plausible content that doesn't match the input context.

Beyond the basic GAN setup, some advanced generative models have influenced inpainting:

- **Conditional GANs with Priors:** Yeh *et al.* (2017) approached inpainting by finding latent vectors of a generative model (like a pre-trained GAN on full images) that, when projected to image space, agree with the known pixels. This *GAN inversion* perspective treats inpainting as an optimization in the latent space of a generator, leveraging powerful pre-trained generative models. It is computationally heavy but showed that a strong prior (like a GAN trained on faces) can fill large missing regions in a plausible way.

- **High-Resolution Inpainting GANs:** More recent models like Zhao *et al.* 2021, *Co-ModGAN* integrate StyleGAN-like architectures for inpainting. Co-ModGAN introduced a co-modulated generator, blending the benefits of an unconditional GAN (which can produce high-res details via style-based modulation) with conditioning on the input image. Such architectures manage to inpaint very large holes (e.g. most of a face) at resolutions up to 1024×1024 , producing results that often surpass previous methods in fidelity. They achieve this by injecting noise and style at multiple scales, giving the generator stochasticity to produce multiple plausible outcomes and fine detail, while using a strong network backbone to maintain coherence with known regions.
- **Multi-modal Output and Diversity:** Traditional GAN training often yields one deterministic output for a given input (since the generator is a function without explicit noise input in many inpainting setups). Some approaches incorporate noise or latent codes to allow diverse inpainting results for the same input mask, addressing the one-to-many nature of the task (many plausible ways to fill a blank). For example, PD-GAN (2021) and other *diverse inpainting* works use conditional GANs that sample different fills. While this is a bit tangential to the main focus, it's worth noting as a trend made easier by GAN frameworks.

In essence, GANs provide the learning framework to judge realism, which is hard to capture with simple pixel-wise losses. However, GANs alone do not inherently solve the issue of how to propagate context information from distant parts of the image into the hole. This is where **attention mechanisms and Transformers** come into play in recent models, as they can better capture global relationships. Next, we discuss Transformer architectures and their role in vision tasks, especially for capturing long-range context in images.

Vision Transformers and Self-Attention in Inpainting

Transformers, originally introduced for sequence modeling in NLP (Vaswani *et al.*, 2017), use multi-head self-attention layers to allow every element (word or image patch) to attend to every other element. When applied to images, **Vision Transformers (ViT)** (Dosovitskiy *et al.* 2020) demonstrated that pure transformer architectures (treating the image as a sequence of patches) can achieve excellent results on image classification, given sufficient data. The core strength of transformers is in modeling long-range dependencies: each self-attention layer blends information globally, unconstrained by a limited receptive field as in CNNs.

In the context of image inpainting, the ability to capture **global context** is extremely valuable. For example, consider an image with two disconnected visible regions separated by a large hole. A CNN might have trouble relating those two regions without many layers or dilations, but a transformer could directly attend between pixels on opposite sides of the hole, ensuring consistency. Transformers, however, come with challenges:

- **Computational Cost:** Self-attention has quadratic complexity in the number of tokens (pixels or patches). For high-resolution images, a naive transformer over all pixels would be infeasible. This is why many vision transformers either work on lower-

resolution feature maps or use patch embedding (e.g., 16x16 patches as tokens) to reduce sequence length. In inpainting, some methods restrict attention to a coarse stage or employ sparse attention patterns to limit computation.

- **Data Requirements and Training Difficulty:** Transformers generally require large training datasets to learn inductive biases (unlike CNNs which have built-in locality bias). In practice, many inpainting approaches therefore combine transformers with convolutional layers to get the best of both worlds.

Several approaches have integrated self-attention into CNN-based inpainting networks even before full transformers became popular. For instance, the contextual attention of Yu *et al.* (2018) can be seen as a form of attention mechanism specialized for matching patches. Later, **Attention-GAN** models and **Dual Attention** modules appeared in image completion literature, allowing the network to explicitly copy or reference distant features. These laid the groundwork for using a true transformer architecture for inpainting.

A **Vision Transformer for generation** divides the image (or its representation) into tokens and uses self-attention to model their relationships. In inpainting-specific transformers, a typical strategy is:

- *Encode the known pixels:* The visible parts of the image (and perhaps a mask indicating missing parts) are fed into some encoder (could be convolutional or a transformer encoder) to produce a set of latent tokens or feature maps.
- *Use a transformer to propagate information:* The transformer can operate either in the image pixel space or in a latent feature space. For example, one could flatten a low-resolution feature map of the image and perform self-attention, thereby allowing information to flow from known to unknown regions. Some works create a token for each known patch and perhaps initialize tokens for missing patches as well, then use transformer layers to fill in the missing content (the missing tokens get values through attention from known tokens).
- *Decode to image:* A decoder (transformer-based or convolutional) then produces the output image, merging the filled content with the original.

The simplest use of transformers in inpainting is to insert a **self-attention layer in a U-Net** at bottleneck or high-level feature layers. This was done for example in some 2019–2020 models, where authors noted improved coherence for large holes if the network can attend globally at least in one layer. But more recent works take this further, designing dedicated transformer components:

- The Mask-Aware Transformer (MAT) by Li *et al.* 2022 uses a transformer architecture that is *aware of the mask*. It restricts attention to only valid tokens (i.e., it prevents meaningless attention from missing pixels to other missing pixels) to reduce computation and focus on connecting known regions to unknown ones. This mask-gating in attention ensures efficiency while still capturing long-range info.

- Dong *et al.* 2022, *ICT - Incremental Transformer for Structure* and its extended version ZITS (Zero-initialized residual incremental transformer, 2023) approach inpainting by splitting the problem into structure and texture. They use a transformer-based network to complete a *structural representation* (like an edge or line drawing of the image) at a lower resolution, then use a CNN to finalize texture details. The transformer in the structural stage can handle global geometry completion (e.g., completing the outline of an object through the hole), which is then used to guide the second stage.
- Some methods incorporate pre-trained vision transformer models. For example, a 2024 approach (Du *et al.*, *InViT*) leverages a transformer trained via GAN inversion: it uses the latent space of a generative model in combination with a ViT to infer masked content, showing a novel way to blend transformer representation power with a GAN prior.

Importantly, **hybrid architectures** that combine CNNs and transformers have become prominent because they leverage CNNs' efficiency and detail-generating capacity with transformers' context modeling:

- The ITrans (Miao *et al.* 2024) is a good representative example. It augments a traditional encoder-decoder (UNet-like) CNN with two transformers: a *global transformer* block that operates on the higher-level features to capture relationships across the whole image, and a *local transformer* block that operates on lower-level features in a restricted neighborhood to refine details. By injecting these blocks, ITrans can maintain fine details (thanks to CNN and local attention) while ensuring global coherence (thanks to the global attention path). The result is an end-to-end network that outperformed many prior purely-convolutional methods on benchmarks.
- Chen *et al.* 2024, *HINT* similarly presents a transformer-based inpainting model that introduces a mask-aware downsampling to preserve as much information as possible from known regions when reducing spatial size (avoiding losing details of the known context). HINT's transformer layers are optimized to handle large missing areas, using efficient attention and sandwiching feed-forward layers to strengthen the transformer. By combining these with convolutional refinements, HINT achieves high quality on multiple datasets (including very challenging ones like ancient wall paintings as discussed later).
- Some research also explores **auto-regressive transformers** for inpainting, treating it as a sequence generation problem (pixel by pixel or patch by patch). For example, a model might generate pixels row by row, conditioned on both left context and any available context from unmasked regions. These approaches, akin to image GPT or diffusion models, can produce very accurate results but tend to be slower than the above parallel methods.

Transformers have also been extended to **video inpainting**, where attention can be applied in the temporal dimension to maintain consistency across frames (though that is beyond our current scope, it's an emerging area building on image methods).

To summarize, vision transformers provide mechanisms to:

- **See the Big Picture:** An inpainting network with a transformer can infer that two edges on opposite sides of a hole should meet, or that the texture/color on one side of the hole should propagate to the other, even if the gap is large.
- **Impose High-Level Consistency:** Self-attention can implicitly enforce things like symmetry, object coherence (e.g., if half a car is visible, it can help complete the other half properly), and avoid contradictions that CNNs might miss if parts of the hole are filled independently.
- **Integrate with CNNs:** Most practical designs use transformers at bottleneck or high-level layers and CNNs at the front/back ends. This yields a hybrid with manageable complexity and training needs.

Having covered GANs and transformers separately, we now focus on **hybrid models** that explicitly combine both: using transformers inside a GAN framework for image inpainting. These models aim to harness transformers' context understanding with the powerful texture synthesis that GANs offer.

Hybrid Transformer–GAN Architectures for Inpainting

Recent state-of-the-art inpainting systems often employ hybrid architectures, where transformers and GAN components complement each other. Below, we outline notable examples of Transformer–GAN hybrid models and their strategies:

- **MAT (Mask-Aware Transformer, 2022):** MAT by Li *et al.* is a transformer-based generator with mask-aware attention, trained with GAN losses. The architecture is optimized for high-resolution images and large holes. By only attending to valid tokens in the transformer, MAT can be applied on 512×512 images efficiently. It produces diverse outputs by sampling, which is facilitated by the GAN training. The authors report state-of-the-art performance on benchmarks like Places2 and CelebA-HQ, and they provide a reference implementation (see GitHub links in resources).
- **ZITS (2022) and ZITS++ (2023):** The ZITS framework (Dong *et al.*) explicitly splits the problem into structure and texture. It first uses a Transformer-based structure restorer (with a masking positional encoding to handle the hole) to generate a coherent sketch (outlines/edges) of the missing content. Then a Fourier CNN texture generator fills in the actual pixels, guided by the completed structure. A GAN loss (often a multi-scale PatchGAN) is applied on the final output to ensure realism. By modularizing this way, the transformer can focus on global structure without getting distracted by texture details, which are handled in the second stage. ZITS++ further improves the transformer component and training scheme to refine structure prediction. These models particularly excel at scenarios where maintaining geometric consistency is crucial (e.g., completing a half-missing building facade with all windows aligned).

- **Co-Modulated GAN (2021):** Though not a transformer in the generator, CoModGAN's discriminator uses a form of *spatial conditioning*. It is a strong GAN-based inpainting model that can be considered a baseline for hybrid comparisons. The generator is built like StyleGAN2 but conditionally modulated by the known region. Some later variants have explored adding attention layers in this architecture to further improve it. CoModGAN's impressive results (especially on face inpainting with FFHQ) set a high bar that many transformer-augmented models aim to surpass.
- **ICT / HiFill with Transformers (2021):** ICT (High-Fidelity Image Completion with Transformers) and a method called HiFill both incorporated transformers or attention in a GAN framework. HiFill (Yi *et al.* 2020) used very large contextual attention with a multi-scale GAN to handle 1K resolution images. ICT (Zeng *et al.* 2021) introduced a coarse-to-fine generator where the coarse stage used a transformer block to ensure pluralistic (diverse) results. Both approaches confirmed that injecting transformer-based context modeling into GAN pipelines improves quality for large missing regions.
- **TransInpaint (2023):** Shamsolmoali *et al.* 2023 present TransInpaint, which is a generator combining a *context-adaptive transformer* and a *texture enhancement network*. The transformer focuses on ensuring the filled content is semantically consistent with the context (an adaptive attention mechanism based on mask position and visible content), and then a second network merges the coarse output with skip features from earlier layers to add realism. The model is trained adversarially. This hybrid showed improvements on challenging cases with complex scenes or multiple objects, where previous methods like MAT might still blur fine details of objects.
- **HINT (2024):** As mentioned earlier, HINT uses mask-aware encoding and a transformer with an enhanced attention design. It also includes an **adversarial loss** as part of its training (on top of reconstruction, perceptual, and style losses). Thus, HINT's generator is a transformer-based network but it learns under a GAN framework. The authors highlight that without adversarial training, the results lack sharpness and finer textures. HINT's hybrid nature allowed it to top quantitative metrics like PSNR/SSIM as well as perceptual metrics like FID on several datasets.
- **Other Notable Hybrids:** Some works have tried transformers in the discriminator instead of or in addition to the generator. For example, a transformer-based discriminator could evaluate global image consistency more intelligently than a CNN. However, this is less common due to the heavy computation and the fact that the generator is where capturing context matters most. There's also research into combining transformers with *other* generative paradigms for inpainting, such as diffusion models (which inherently have global awareness due to iterative refinement). While diffusion models are beyond our scope here, they represent another modern approach competing with GANs/Transformers in inpainting (e.g., Stable Diffusion can do inpainting guided by text prompts).

In conclusion, hybrid Transformer–GAN models are designed to utilize:

- **Transformers for global semantic understanding:** ensuring the completed image makes sense as a whole.

- **GANs for photorealistic detail:** using discriminator feedback to correct any blurry or inconsistent textures.

These models tend to be complex, often consisting of multi-module architectures, but they currently deliver the best results in inpainting benchmarks. Next, we will discuss how these models are trained, what loss functions and strategies they use, and how we evaluate their performance.

Loss Functions, Training Strategies, and Evaluation Metrics

Training an image inpainting model involves balancing multiple objectives to ensure the output is both correct in content and high in visual quality. Below we outline the common **loss functions**, training techniques, and evaluation metrics used in the literature, especially focusing on those relevant for Transformer-GAN hybrids:

Loss Functions for Inpainting

Most models optimize a weighted sum of several loss terms:

- **Reconstruction Loss:** Typically an L_1 loss (mean absolute error) between the output and ground truth on the masked pixels. Some also apply L_1 or L_2 on the known region to ensure the model doesn't alter unmasked content (identity loss). L_1 is preferred over L_2 as it tends to produce sharper results (less blurring).
- **Adversarial Loss:** As discussed, a GAN loss where the generator tries to fool a discriminator that judges completed images vs. real images. Many inpainting works use a hinge or least-squares GAN loss for stability. The adversarial loss pushes the output distribution of filled pixels towards that of real images. In practice, the adversarial loss might be applied over the whole image or just the filled region (some discriminators are fed with the concatenation of the masked input and the output to explicitly focus on the hole).
- **Perceptual Loss:** This is a loss computed in a deep feature space (e.g. using a pre-trained VGG network on ImageNet). By comparing the high-level feature representations of the output and ground truth, the model is encouraged to produce results that are perceptually similar to real images, not just pixel-wise similar. This helps maintain texture and avoids the *color drifts* or slight blurriness that L_1 alone might allow. Johnson *et al.* (2016) popularized this technique for style transfer and super-resolution, and it has become a staple in inpainting as well.
- **Style Loss:** Sometimes used in conjunction with perceptual loss, style loss (based on Gram matrices of VGG features) ensures the texture statistics (like patch co-occurrences) of the filled region match those of the real image. This was more common in older methods or when doing style-oriented inpainting.

- **Total Variation (TV) Loss:** A regularizer to encourage smoothness in the filled area, reducing noise or checkerboard artifacts. It penalizes high-frequency noise by encouraging neighboring pixels to have similar values.
- **Auxiliary Losses:** For models that predict intermediate representations, losses on those predictions are used. For example, EdgeConnect had an edge prediction loss (to ensure the edge generator produced correct edges). Similarly, if a model like ZITS predicts an edge map or segmentation as a prior, one would include a loss comparing that to the ground-truth edge/structure (if available, or can be derived from the image).
- **Feature Matching Loss:** Used in some GAN setups (especially with multi-scale discriminators) where the generator is additionally tasked to match the intermediate feature representations in the discriminator between real and fake. This can stabilize training and also improve perceptual quality (by implicitly optimizing a perceptual-like metric).

In transformer-based models, these losses remain largely the same. One addition is that some transformers utilize a **mask token reconstruction loss**, essentially treating it like a Masked Image Modeling problem (analogous to BERT's masked word prediction). For example, a transformer might be tasked to directly predict missing pixel values and be evaluated on them with L_1 loss.

Training Strategies and Techniques

Training a hybrid model with many components can be tricky. Common strategies include:

- **Progressive Training:** Start with smaller holes and gradually increase the mask size difficulty once the model learns to handle simpler cases. This curriculum learning can lead to more stable convergence for very large masks.
- **Two-Stage Training:** For multi-stage models (coarse and refine networks, or structure then texture networks), one may first train the first stage alone (e.g., train the structure prediction transformer with a reconstruction loss on edges/structure), then freeze it and train the second stage to generate final output, and finally fine-tune end-to-end with lower learning rate. This ensures each stage does its job well before coupling them.
- **Pre-training and Fine-tuning:** In some cases, using large datasets to pre-train part of the network (for example, a transformer on ImageNet for masked autoencoding, or using a pretrained ViT as initialization) can give a boost. Then the model is fine-tuned on the specific inpainting task/dataset. However, due to differences in data distribution (faces vs. scenes, etc.), many works train from scratch on target data but benefit from large-scale datasets.
- **Mask Schedules:** During training, one typically uses random masks. These can be either random free-form scribbles, irregular holes (perhaps generated by some procedural algorithm as in Liu 2018), or fixed datasets of masks (like the *irregular*

mask dataset provided by Liu *et al.*). A balanced training will expose the model to a variety of mask shapes and sizes. Some authors progressively increase hole size, others randomize from the start. Ensuring the model sees both small and very large missing regions is important so it doesn't overfit to one regime.

- **Stabilizing GAN Training:** Techniques like feature matching loss (as mentioned), using spectral normalization in the discriminator (common in many image inpainting GANs to regularize the discriminator), and iterative training (several generator steps per discriminator step or vice versa) are often used to get stable results. Transformer generators don't inherently destabilize GAN training, but their high capacity means careful tuning of learning rates may be needed to avoid the generator overwhelming the discriminator too quickly.
- **Data Augmentation:** Interestingly, besides random masks, augmentation of the known parts of the image (like random color jitter or affine transforms) is usually not done, because the input image context is sacrosanct. However, augmenting the entire image and mask together (e.g., rotations, flips) is fair game and can effectively increase data variation.

Evaluation of inpainting is multi-faceted:

Evaluation Metrics

- **Pixel-wise Metrics (PSNR, SSIM):** Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) are classical metrics to measure fidelity of the completed image compared to ground truth. PSNR is essentially based on MSE; SSIM measures structural similarity in patches (accounting for luminance, contrast, structure). High PSNR/SSIM indicate the output is very close to the original image pixels. These metrics are useful but they tend to reward safe, blurry predictions over perceptually good ones (a perfectly sharp but slightly incorrect texture might get worse PSNR than a blurry average).
- **Perceptual Metrics (LPIPS):** Learned Perceptual Image Patch Similarity (LPIPS) is a metric that computes the distance between features of the output and ground truth in a deep network (like a trained VGG). Lower LPIPS means the output looks more like the ground truth in a human-perceptual sense. This is more aligned with visual quality than PSNR. In many recent papers, LPIPS and FID (see below) are preferred for comparing inpainting results.
- **FID (Fréchet Inception Distance):** FID computes the distance between the distribution of deep features (Inception network activations) of completed images and that of real images. A lower FID means the set of outputs is statistically closer to real images (captures realism and diversity). It's commonly reported for inpainting models now, especially when evaluating how real outputs look regardless of the specific ground truth.
- **Human Study / MOS:** Many papers include a user study or Mean Opinion Score evaluation where human raters choose which of two results looks more realistic or

give a rating to outputs. This is the ultimate test for perceptual quality. The best models tend to score well here, often with differences that align with FID or LPIPS.

- **Specialized Metrics:** If focusing on certain domains, there can be domain-specific metrics. For example, for face inpainting, one might run a face recognition model to see if identity is preserved (to ensure the inpainting didn't alter the person's identity). Or for text removal in images, OCR accuracy on the region could be measured before and after. For our general summary, these are less common, but it's worth noting that application-driven metrics might be considered in specific projects.

In evaluation, one typically uses a masked region and compares the filled result to the original image (which serves as ground truth for that region). Note that some tasks allow multiple possible fills (especially if the mask is huge, there could be inherent ambiguity). Metrics like FID can account for distributional similarity in such cases, while PSNR cannot.

Transformer-GAN models often report superior **perceptual metrics** (FID, LPIPS) and competitive or slightly better PSNR/SSIM than previous methods. This indicates they are managing to not just copy or blur content but truly generate realistic details that sometimes differ from the exact original pixels but still look plausible (hence slightly lower PSNR, but much better FID).

In summary, a combination of losses is used to train these models (to balance realism and accuracy), and a variety of metrics are used to judge their performance. When developing an inpainting project, it's important to decide which metrics align with your end goals (e.g., for an application like photo editing, perceptual quality is usually paramount).

Application Domains and Current Research Trends

Image inpainting has a wide range of real-world applications. Here we outline several prominent application domains, and then discuss the trends in current research that span across these domains:

Application Domains

- **Photograph Object Removal:** One of the most common uses of inpainting is to remove unwanted objects or people from photographs. For example, removing a stray pedestrian from a scenic shot, or erasing wires and poles from a landscape. In this scenario, the model needs to fill in background texture (foliage, sky, wall, etc.) in a realistic way. Datasets like Places2 or COCO (with segmentation masks of objects) are often used to simulate this by cutting out objects. Successful methods enable "virtual photo editing" where users can brush out content and get a natural result as if it was never there.
- **Face Image Completion & Editing:** Inpainting is heavily used for face images, for tasks like restoring old portraits (where parts are damaged or scratched), removing blemishes or obstructions (like a microphone covering part of a speaker's face), or even transforming facial attributes by removing and regenerating certain regions (for

instance, removing glasses or changing hairstyle by repainting those pixels). Models like those trained on CelebA-HQ or FFHQ specialize in faces and can produce remarkably coherent eyes, noses, mouths in the masked regions. There's also interest in privacy-related face editing: e.g., removing identifying features and inpainting to anonymize faces.

- **Photo Restoration (Damage Repair):** Beyond faces, any old or deteriorated photograph or artwork can benefit from inpainting to fill cracks, missing patches, or faded areas. This often requires the model to hallucinate plausible content that matches the original style (greyscale film grain, or oil painting texture). Inpainting models might be combined with scratch detection or segmentation to specifically target damaged regions.
- **Art and Cultural Heritage:** A fascinating domain is restoring ancient paintings or frescoes where paint has chipped off. The *Dunhuang Grottoes* challenge mentioned earlier is an example: these are ancient murals in caves where large parts may be missing. An inpainting model here has to continue patterns like ancient clothing or scenery that might only be partly visible. This is challenging due to unique artistic styles and the need for historical authenticity. Some methods integrate priors about symmetry or use references from undamaged parts of the artwork.
- **Graphics and Creative Applications:** Inpainting can also be used creatively in graphics design, such as filling in textures on 3D models from partial textures, or completing user sketches (sketch → realistic image, where the input can be seen as an extreme mask). Tools like Adobe Photoshop have incorporated content-aware fill (which is a form of inpainting). Additionally, in video games or VR, inpainting can help fill in missing view areas (like turning a panorama into a full 360-degree view by inpainting the nadir).
- **Video Inpainting:** Extending image inpainting to video involves removing objects or restoring frames in a video while keeping temporal consistency. Applications include film post-production (removing wires, logos), surveillance video editing, or error concealment in video transmission. While it's an extension of image inpainting, specialized models are needed to handle the time dimension (ensuring that the inpainted object does not flicker or jump between frames). Some transformer-based approaches have started to tackle video inpainting by attending across time as well as space.
- **Medical Imaging:** In medical images, inpainting can be used to reconstruct missing or corrupted portions of scans. For example, removing an artifact from an MRI and filling it with plausible tissue texture, or inpainting could be part of data augmentation (deliberately mask parts of an image and train a model to predict them, thereby forcing the model to understand global structure of, say, an organ). Care is needed, of course, as hallucinated medical content should not mislead diagnoses—so this is more for assisting or restoring data rather than final diagnostic images.

- **Multi-modal Inpainting:** A recent trend is guiding inpainting with additional inputs: e.g., *text-guided inpainting* (like using a prompt "fill this hole with a tree" for generative models), or using a reference image to borrow style from. While not the focus here, it's an exciting application that combines vision and language.

Given the user hasn't decided on a specific domain, it's worth noting that many of the models we've discussed can be fine-tuned or adapted to any of these domains. Typically, the difference lies in training data: - For faces, one uses face datasets (CelebA-HQ, FFHQ). - For general scenes and object removal, one uses Places2 or COCO. - For art restoration, one might need a specialized dataset (like Dunhuang or other artwork images). - Some models can generalize across domains fairly well (especially if trained on large diverse data), but best results come from domain-specific training.

Current Research Trends

The field of image inpainting is actively evolving. Key trends in the past 5–6 years include:

- **Transformer-Based Methods:** As extensively discussed, applying transformers is a major trend. A recent survey by Elharrouss *et al.* 2023 reviews these techniques, highlighting that transformers improve handling of large holes and complex scenes. Many top methods in 2022–2024 (MAT, ZITS, HINT, etc.) use transformer components.
- **Diffusion Models and Autoregressive Models:** Beyond GANs, diffusion models (like latent diffusion or Denoising Diffusion Implicit Models) have been applied to inpainting. Notably, Stable Diffusion (Rombach *et al.* 2022) can be used for inpainting by conditioning on masked images and a text prompt. These models can produce very diverse and high-quality outputs and have garnered attention for tasks where you might want creative or user-guided fills. However, they are often larger and slower to run than specialized inpainting networks. The line between pure AI art and inpainting is blurring with such tools, but in research, diffusion-based inpainting is likely to be an important direction.
- **Higher Resolutions and Efficiency:** Early deep inpainting papers often worked at 256×256. Now, models like LaMa and CoModGAN comfortably handle 512 or 1024 resolution. There's a push for even higher (4K images, for professional photo editing). This introduces memory and efficiency challenges. Researchers are exploring more efficient attention (like linear attention or localized attention windows as in Swin Transformer) and better memory management (multi-stage inference, tiling large images). One trend is using multi-resolution training: train a model to work at variable resolutions so that at test time it can adapt (LaMa touched on resolution robustness with Fourier convolutions).
- **Integration of Structure Guidance:** We see a lot of methods explicitly incorporating structure: edges (EdgeConnect), segmentation maps (some works condition on scene segmentation to ensure logical fills, e.g., sky region gets sky texture), or depth maps (for completing images in 3D-aware ways). Using such side information can greatly help in specific scenarios. For example, a street scene inpainting might benefit from a layout parser that guides the network not to place a tree in the middle of a road hole.

This trend reflects a move towards more *semantic understanding* within inpainting models.

- **Evaluation and Benchmarks:** There's growing recognition that traditional metrics like SSIM are not sufficient. Thus, new benchmarks often include perceptual user studies. Also, there are now standardized benchmarks (like Places2, CelebA-HQ, Paris StreetView) with held-out masks for fair comparison. The community also releases *challenge datasets* (like Dunhuang or the more general Paris StreetView challenge in the past) to push models to their limits. Evaluating on extremely large holes (80-90% of image missing) has become a way to differentiate model capabilities.
- **Generalization and Unsupervised Inpainting:** Most models are supervised, but a trend is semi-supervised or unsupervised inpainting. For instance, using generative models pre-trained on unmasked images (like GANs or masked autoencoders) and then fine-tuning for inpainting with limited paired data. The GAN inversion approach cited is one such attempt. Masked image modeling (like MAE by He et al. 2022) itself is essentially an inpainting task used for pre-training. Some research explores how an MAE-trained ViT can be directly used for inpainting by transferring its encoder-decoder weights.
- **User Interaction and Control:** There's also an increasing interest in giving users control over the inpainting result. E.g., interactive inpainting where a user can sketch a hint of what should be in the hole (and the model completes it following that hint), or choose from multiple proposals. Some hybrid models allow sampling different outcomes (due to random latent or noise inputs, or explicitly modeling uncertainty). This is important because for truly large holes, there isn't a single "correct" answer.

In summary, image inpainting is not a solved problem, but the combination of transformers and GANs has substantially closed the gap between automated inpainting and what a skilled human retoucher could do. The choice of approach may also depend on the application: for realistic photograph editing, one might pick a method that is conservative and consistent (leaning more on reconstruction loss + minor adversarial); for creative tasks, one might use a method that allows more diversity (e.g., with stochastic sampling or a diffusion model).

The field is moving towards models that are *general and flexible* yet deliver *high fidelity*, and the Transformer-GAN hybrids are a key step in that direction.

Datasets for Image Inpainting

A variety of datasets are used in image inpainting research, each serving different domains and challenges. Below is a list of key datasets, including their domains, typical resolution, and common uses. Links to their sources are provided for further details or downloads:

- **Places2 Scene Dataset** – [Places: An Image Database for Deep Scene Understanding \(Zhou et al. 2017\)](#). *Domain:* Diverse natural and urban scenes, landscapes, indoor rooms, etc., across 365 categories. *Scale:* Over 8 million training images; often resized

to 256×256 for inpainting model training. Higher-resolution versions available (images can be up to 512 or 1024 on the longest side). *Use:* A primary dataset for general-purpose inpainting. Models trained on Places2 can handle wild nature scenes, cityscapes, and more. The diversity helps networks generalize. It's commonly used for object removal or general hole-filling tasks. (Places2 is a successor to the original Places and is used in the Nvidia inpainting paper and others.)

- **CelebA & CelebA-HQ (Faces)** – [CelebA Dataset Page \(Liu et al. 2015\)](#). *Domain:* Celebrity face images. CelebA has 202,599 face photos of 10,177 individuals, with varied poses and backgrounds (size 178×218, often cropped to square 128 or 256 for training). CelebA-HQ is a high-quality subset of 30,000 images at 1024×1024 resolution with better visual fidelity (created by Karras et al. 2018 via an improvement procedure). *Use:* Primarily for face inpainting tasks. Good for evaluating if a model can realistically regenerate eyes, nose, mouth etc., and maintain identity or at least consistency. Many face-focused inpainting models (and editing models) report results on CelebA or CelebA-HQ.
- **FFHQ (Flickr-Faces HQ)** – [FFHQ Dataset in StyleGAN paper \(Karras et al. 2019\)](#). *Domain:* High-resolution human faces from Flickr, with great diversity in age, ethnicity, lighting, accessories, etc. 70,000 images at 1024×1024. *Use:* Advanced face inpainting and generative models. Co-ModGAN, for instance, trains on FFHQ for face completion. FFHQ's diversity challenges models to handle a wide range of face attributes. Often used when the goal is photorealism at high resolution. (Note: License is non-commercial, researchers use it for model development and evaluation.)
- **Paris StreetView** – Introduced in [Pathak et al. 2016 \(Context Encoders\)](#). *Domain:* Street scenes from Paris, particularly focusing on building facades, streets, and windows. *Scale:* 15,000 images (the exact number not publicly confirmed due to dataset not released openly). Images are typically used at 256×256. *Use:* Historically a benchmark for inpainting (especially structure completion like building architecture). Many papers show qualitative results on Paris StreetView (e.g., removing vehicles from street or reconstructing parts of buildings). Because of legal reasons (Google Street View data), the dataset isn't freely downloadable; researchers often obtained it privately. Nevertheless, results on it demonstrate a model's ability to handle perspective and repeating patterns (windows, balconies). Some later works opt for other cityscape sets or use Places to cover similar content.
- **ImageNet** – [ImageNet ILSVRC dataset \(Deng et al. 2009\)](#). *Domain:* A very large collection of images across 1000 object categories. While ImageNet is designed for classification, it has been used in inpainting research (Pathak et al. trained context encoders on ImageNet in an unsupervised way). *Scale:* 1.3 million images of varied resolution (often downsampled to 128 or 256 for experiments). *Use:* If a model can be trained on ImageNet, it indicates robustness across many object types. Often, researchers will train on a subset or use ImageNet to pretrain and then fine-tune on a more specific dataset. Not a focused inpainting benchmark for quality, but good for representation learning.

- **MS COCO** – [COCO: Common Objects in Context \(Lin et al. 2014\)](#). *Domain*: Everyday scenes with multiple objects, annotated with segmentation masks and bounding boxes. 120k training images. *Use*: COCO is used to evaluate object removal specifically: one can take an object segmentation mask (say a person or an animal) and remove that region, then inpaint it. The dataset provides realistic contexts and known ground truth for the removed object region. Some papers generate random irregular masks on COCO and report scores, or use COCO's things vs. stuff segmentation to remove "things" and inpaint using surrounding "stuff". It's a challenging test since objects have complex boundaries and the backgrounds can be varied.
- **Dunhuang Grottoes Painting Dataset** – [Yu et al. 2019, Dunhuang Restoration Dataset](#). *Domain*: Artwork (ancient Chinese murals) with damage. *Scale*: A few thousand images (with corresponding masks of damaged regions for a challenge). *Use*: Specifically for evaluating inpainting in an art restoration context. The dataset became available as part of a challenge; it contains high-resolution images of murals with sections missing. Inpainting here requires not just filling space, but doing so in the artistic style of the original (brushstrokes, colors) and often inferring missing semantic content (like parts of a deity's robe or scenery). This is a very different domain than everyday photos, testing a model's adaptability.
- **WikiArt (Paintings) and Behance Artistic Media**: These are art image datasets. WikiArt has many painting images across styles (used in some works for style transfer and potentially inpainting). Behance contains illustration and artwork images. *Use*: Researchers interested in creative inpainting or completing artwork might use these to train or test models. For example, one might try to remove an object from a painting or complete a half-painted artwork. The challenges include dealing with non-photorealistic content, which some transformers might handle better than CNNs due to flexibility in pattern generation.
- **Oxford Flowers, CUB Birds (Object-specific datasets)**: These contain multiple images of specific object classes (flowers, birds) often used in conditional generation. While not common for inpainting benchmarks, one could use them to test if a model knows how to reconstruct a specific type of object when partially masked (e.g., remove half a bird and see if it can regrow it correctly). Typically, general models can be fine-tuned on these for domain-specific inpainting if needed.

In practice, for a new project, choosing a dataset means deciding on the domain of interest and the available data for that domain. Many state-of-the-art models are first trained on Places2 (for generic scenes) or CelebA/FFHQ (for faces). If you aim to build an inpainting model for, say, satellite imagery or medical images, you'd gather a dataset in those domains and could start from a model pre-trained on a broad dataset then fine-tune.

Each dataset above has its own distribution: models can sometimes generalize but often a model trained on one (e.g., Places) will produce slightly blurry or inconsistent results on faces, and vice versa, if not adapted. So, dataset choice is critical.

Frameworks and Implementation Considerations

Implementing advanced inpainting models (especially those with hybrid architectures) requires choosing a suitable deep learning framework and leveraging existing resources.

Preferred Frameworks

The most widely used framework for image inpainting research is **PyTorch**. PyTorch is popular due to its dynamic computation graph (easy debugging and customization), extensive community support, and plethora of open-source implementations of GAN and transformer models. Most of the GitHub projects referenced (see next section) are implemented in PyTorch. It offers convenient libraries for transformations, loss functions (e.g., perceptual loss implementations), and pre-trained models (like VGG for loss or Inception for FID computation). The ecosystem also has PyTorch Lightning and other tools that can structure GAN training nicely.

Another framework is **TensorFlow** (and Keras). Some earlier inpainting works (2016-2018) were in TensorFlow, and one can certainly implement these models in TF2/Keras. However, in recent years PyTorch has overtaken it in research for GANs and transformers. If one's project is more research-oriented or custom, PyTorch is usually the go-to. If there's a need to integrate into certain production environments or use Google Cloud TPUs, TensorFlow might be considered; but even then, many choose to convert PyTorch models via ONNX or other means rather than develop directly in TF.

There is also **JAX** (by Google) which some cutting-edge research uses for its speed in large-scale training (for example, Google's ViT was originally in JAX/Flax). But for image inpainting specifically, JAX is less common, and the lack of many out-of-the-box GAN training utilities means a higher overhead.

In summary, **PyTorch is recommended** for its ease of use and the fact that many pre-trained models and codebases are available to build upon.

Existing Libraries and Tools

Beyond base frameworks, some specialized libraries can accelerate development:

- **MMagic (OpenMMLab)**: MMagic is a library that includes implementations of various image generation and editing models, including inpainting (with some of the known algorithms implemented and ready to use/train). It's built on PyTorch and can save time if it includes the model you need (e.g., they have partial conv inpainting, deepfill, etc.). It also provides a unified interface for evaluating and comparing methods.
- **Hugging Face Transformers / diffusers**: If using a transformer or diffusion approach, HuggingFace provides ready implementations of ViTs and even a pipeline for inpainting with Stable Diffusion. While heavy, these can be adapted for specific tasks and they use PyTorch under the hood.

- **Neural Blocks for GANs:** There are repositories like CompVis Latent Diffusion or StyleGAN2 that can be adapted for inpainting with some coding. StyleGAN2 was adapted by CoModGAN authors for example. Using these as a starting point can ensure a solid baseline for the generative part.
- **Evaluation Code:** Implementations for FID, LPIPS, etc., are available (e.g., LPIPS code by Richard Zhang, and many PyTorch FID implementations). It's recommended to use the same evaluation tools as the literature to compare results accurately.

When implementing a hybrid model, keep an eye on:

- **Memory usage:** Transformers can be memory-hungry. Techniques like gradient checkpointing (trading compute for memory) might be necessary for large images.
- **Distributed training:** If using large datasets like Places2, you might need to train on multiple GPUs. PyTorch's Distributed Data Parallel can help scale the training.
- **Precision:** Some implementations use mixed precision (FP16) for faster training, which is feasible for inpainting but one must ensure losses like adversarial loss still work well (sometimes low precision can destabilize GAN training if not careful).

Finally, consider using **pre-trained weights** if available. For example, if the goal is to implement a known model (like MAT or LaMa) for a baseline, using their released weights to test on your data can set a performance reference. Then you can fine-tune or modify from there.

We now provide references to important papers and resources, as well as links to code repositories that could serve as a starting point or reference for implementation.

References and Further Reading

Below are key references, including foundational papers, recent papers (within ~6 years), blog posts/tutorials, and open-source implementation repositories relevant to image inpainting and particularly to transformer-GAN hybrid approaches. The list is organized by category for clarity:

Foundational Papers and Surveys

- Bertalmio et al., 2000 – *Image Inpainting*. (Seminal work introducing the basic concept of digital inpainting via diffusion of pixel intensities.)
- Criminisi et al., 2004 – *Region Filling and Object Removal by Exemplar-Based Inpainting*. (Classic patch-based inpainting algorithm that propagated texture by sample-and-vote mechanism.)
- Pathak et al., 2016 – *Context Encoders: Feature Learning by Inpainting*. (First deep learning approach using an encoder-decoder CNN and adversarial training to fill missing central regions.)

- Iizuka et al., 2017 – *Globally and Locally Consistent Image Completion*. (Introduced global/local discriminators; a milestone in GAN-based inpainting.)
- Yu et al., 2018 – *Generative Image Inpainting with Contextual Attention*. (Two-stage inpainting with novel attention layer; widely cited and the code is a common baseline.)
- Liu et al., 2018 – *Image Inpainting for Irregular Holes Using Partial Convolutions*. (NVIDIA's partial conv approach; became popular for robust inpainting, with an easy-to-use public model.)
- Yu et al., 2019 – *Free-Form Image Inpainting with Gated Convolution*. (Improved free-form mask inpainting using learnable gates on conv filters; known as DeepFillv2 in some implementations.)
- Nazeri et al., 2019 – *EdgeConnect: Structure Guided Image Inpainting*. (Two-stage edge then image GAN model; code released and used in many comparisons.)
- Suvorov et al., 2022 – *Resolution-Robust Large Mask Inpainting with Fourier Convolutions (LaMa)*. (Introduced fast Fourier convolution to achieve huge receptive fields without attention; a non-transformer approach that set high benchmarks and is very practical. Code available.)
- Elharrouss et al., 2023 – *Transformer-based Image and Video Inpainting: A Review*. (Comprehensive survey of recent inpainting methods, focusing on transformer techniques and providing a taxonomy of approaches and future directions.)

Transformer & Hybrid Inpainting Models (2019–2024)

- Zeng et al., 2021 – *High-Fidelity Pluralistic Image Completion with Transformers (ICT)*. (One of the earlier works to explicitly use transformer layers in an inpainting generator; produced diverse outputs.)
- Zhao et al., 2021 – *Large Scale Image Completion via Co-Modulated GANs (CoModGAN)*. (Though transformer-free, it's a modern GAN model that many hybrids compare against. It uses StyleGAN architecture for inpainting; open-source code by authors.)
- Li et al., 2022 – *MAT: Mask-Aware Transformer for Large Hole Image Inpainting*. (CVPR 2022. Introduces a specialized transformer block for inpainting. Achieved SOTA on multiple benchmarks. **[Code released]**)
- Dong et al., 2022 – *ZITS: Structure Enhanced Image Inpainting with Incremental Transformer*. (CVPR 2022. Two-stage approach with transformer for structure, CNN for texture. **[Code released]**.)
- Li et al., 2022 – *AOT-GAN: Attention-Oriented Transformer for Inpainting*. (IEEE TCSVT 2022. Another variant of combining attention mechanisms in GAN for inpainting.)
- Zhu et al., 2022 – *BAT-Fill: Bidirectional Autoregressive Transformers for Diverse Inpainting*. (ACM MM 2022. Uses autoregressive transformer so it can generate multiple different fills.)

- Miao et al., 2024 – *ITrans: Generative Image Inpainting with Transformers*. (Published in 2024, open access. Proposes global+local transformers in a CNN framework. Good ablation studies on how transformers improve results.)
- Chen et al., 2024 – *HINT: High-Quality Inpainting Transformer*. (Preprint 2024. Focuses on preserving information with mask-aware downsampling, and efficient attention for large holes.)
- Shamsolmoali et al., 2023 – *TransInpaint: Inpainting with Context-Adaptive Transformer*. (ICCV 2023 Workshop. Combines transformer and CNN in a novel generator architecture.)
- Liu et al., 2023 – *CTRF-GAN: Image Inpainting with CSWin Transformer and Fast GAN*. (IEEE Transactions 2023. Uses CSWin (Cross-shaped Window) transformer blocks inside a GAN; shows the trend of trying different transformer variants.)

Tutorials, Blogs, and Informative Posts

- Murat Çelik, 2023 (Medium) – *Brief Summary of Datasets for Image Inpainting*. (Provides plain-language descriptions of commonly used datasets like Paris, Places, etc., and context about their use.)
- Akash Panigrahi, 2019 (TowardsDataScience) – *Image Inpainting with Deep Learning*. (An accessible introduction to deep inpainting, albeit slightly older (pre-transformers), explaining context encoder and partial conv approaches.)
- Two Minute Papers YouTube – *Review of Latest Inpainting Models*. (Short video summaries of some inpainting papers, good for a quick visual understanding of results.)
- Casual GAN Papers – *CoModGAN Explained*. (Blog-style explanation of CoModGAN architecture and results. Useful to grasp how advanced GAN-based inpainting works, which is complementary to transformer-based approaches.)
- PapersWithCode – Image Inpainting Task (Online leaderboard and list of papers with code. Helpful to see SOTA methods and their reported metrics on standard datasets. Often includes direct links to code.)

Code Repositories (GitHub)

- JiahuiYu/generative_inpainting – Official code for Yu et al. 2018 (contextual attention). TensorFlow. A good baseline to see how two-stage models were implemented.
- NVlabs/partialconv – Official PyTorch code for Partial Convolution based inpainting (Liu 2018). Contains training and inference code, and even a training mask dataset. Can be used as a baseline or for comparisons.
- knazeri/edge-connect – Official PyTorch implementation of EdgeConnect (Nazeri 2019). Includes pretrained models and easy inference for structure-guided inpainting.

- [zsyzzsoft/co-mod-gan](#) – Official TensorFlow implementation of CoModGAN (Zhao 2021). High-quality model for large-hole inpainting. Even if not using directly, the architecture and trained weights can be informative.
- [fenglinglwb/MAT](#) – Official PyTorch implementation of MAT (Mask-Aware Transformer, 2022). Includes network code, training procedure, and pretrained weights for Places2 and CelebA-HQ. A valuable resource for understanding how to integrate transformers in a high-res inpainting model.
- [DQiaole/ZITS_inpainting](#) – Official PyTorch code for ZITS (CVPR 2022). It provides the two-stage structure-texture framework. One can run this code to reproduce results or use it as a starting point for improvements.
- [Saafke/EDFlow-Inpainting](#) – Unofficial but clean PyTorch implementation of LaMa (Fourier inpainting) and other methods. Useful to compare non-transformer approach with transformer ones.
- [chenShuang15/HINT](#) – (If available) would be the code for HINT. [At the time of writing, one would check the authors' page for a code release link, as it's very new.]
- [AlonzoLeeoooo/awesome-image-inpainting-studies](#) – An *Awesome List* of image inpainting papers and codes. Great for finding specific implementations or for keeping track of the newest papers in this field.

This concludes the comprehensive overview. With the above breakdown of topics, one should have a solid understanding of the principles behind image inpainting, the latest architectures (especially those merging transformers and GANs), how to train and evaluate such models, the various domains in which they can be applied, and resources to dive deeper or get hands-on with code and data. Equipped with this knowledge, you can proceed to identify a target application domain for your project (be it face restoration, object removal in scenes, art restoration, etc.) and select or develop a suitable Transformer-GAN hybrid model using a framework like PyTorch.